

Wenjie LUO 罗文杰

Agentic AI product & evaluation · Transport engineering (NUS MSc)

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GitHub: github.com/LUOaini1213 | Site: luoini1213.github.io (all projects open source)

EDUCATION

National University of Singapore (NUS) MSc Civil Engineering · Transportation & Urban Transport Aug 2025 – Jan 2027

- Intelligent transport & simulation. Coursework: Machine Learning Foundations (CEE5901QA), Advanced AI Methods (CEE5901QB); dissertation CE5001, coursework CE5212 (transport AI).

Chang'an University (211) BEng Transportation, Excellence Engineer Programme

Sep 2021 – Jun 2025

- Top ~15% of cohort. Coursework includes Advanced Mathematics I (93), Computing (90), Python programming, engineering economics.

PROJECTS

Civil Buddy open-source agentic AI workspace · author

Jul 2026 – present

- An agent workspace ("Codex for civil engineering"): **66** role skills in **16** categories run through SOPs; hard numbers come from deterministic engines — the model never writes coordinates or container counts; high-risk writes require human confirmation.
- 128** automated packing evaluations (16 lanes × 8 rounds, rerunnable script); same-order comparison: LLM self-planning **29** containers vs deterministic engine **25** — hard numbers must come from tools.
- 200+** commits, CI smoke tests, trial packages on GitHub Releases; one-page PRD with acceptance criteria; refusals carry reasons, retries degrade through an audit trail, cost circuit-breaker.

NAVSIM ability ladder — what caps open-loop planning open source · self-study

Aug – Sep 2026

- Built a **10**-agent ladder (ConstantVelocity **0.233** → Human **0.945**) on **563** scenes and one metric cache; paired bootstrap 95% CI on every rung.
- Kinematics first: **0.233** → **0.580**, NC failures 39% → 12%. GT boxes do not close the gap; the map does (DAC 0.737 → **0.950**).
- Learned controls: a model asked to draw the whole line scores **0.527** (hand rule **0.730**); let the rule draw the line and the model only pace it and it reaches **0.763** — a learned part earns its place only in the right slot.
- 62 of the 64 mini logs are the eval logs, so learned scores are contaminated; re-scored all agents on **135** held-out scenes and downgraded what stopped separating.

NUS CE5212 traffic coordination individual project

Mar – Apr 2026

- "Rules propose, the LLM only approves": **43** of 44 candidate actions approved correctly (**97.7%**, course log) — more reliable than letting the model decide directly.
- Two failure findings: direct LLM decisions hallucinate state; synchronous calls slow the simulation (Bus 95 delay +2.2 min) — approval belongs async / in shadow mode.

NUS CE5203 AYE weaving-bottleneck study group of 4

Mar – Apr 2026

- YOLOv11 video counts calibrated a SUMO network; compared variable speed limits and ramp metering across **12** scenarios; delivered mitigation recommendations.

NUS CE5001 flood resilience of the bus network dissertation

Jan – Apr 2026

- ~**3,500** lines of Python / 10 modules; **793** bus services, **5,201** stops; **200** paired rule-vs-model experiments; on dense networks the rule-based plan cut relevant travel time by **62.2%**.

EXPERIENCE

Far East Facade (Singapore) Project Engineer

Jun 2026 – present

- Track **15** façade material types and inbound plans on a kanban across **5** suppliers; a Python tool parses supplier shipment / test emails into arrival batches and inspection results, replacing manual checking.

Chery International Product Manager Intern

May – Jun 2026

- Competitive win/loss and VOC mining for overseas markets — **15** user personas, **5** VOC reports; bubble-chart market positioning (length / price / registrations) with a sales-trend forecast.

CaiXun Group (Colorful-Clouds) Product Manager Intern

Jun – Jul 2025

- Wrote **3** PRDs (features, interaction logic, acceptance criteria), all approved; built **5** Axure core-page prototypes that entered the development schedule.

Expeditors Supply-Chain Assistant

Dec 2024 – May 2025

- Coordinated carriers, clients and customs brokers in an English-speaking environment; issued and reviewed about **50** customs declarations per day.

SKILLS & OTHER

- Product & agents:** PRD, Axure, acceptance criteria; agent specs (tool calls, error handling); competitive win/loss, VOC, personas.
- AI & evaluation:** NAVSIM PDMS comparison and failure attribution; shadow evaluation; tools compute, the LLM orchestrates.
- Engineering & languages:** Python, SUMO simulation, YOLOv11 counting, Excel; English-medium MSc; English working environment in Singapore.
- Competitions & campus:** TikTok TechJam 2026 — four-track submissions (results pending); Youth League branch secretary.